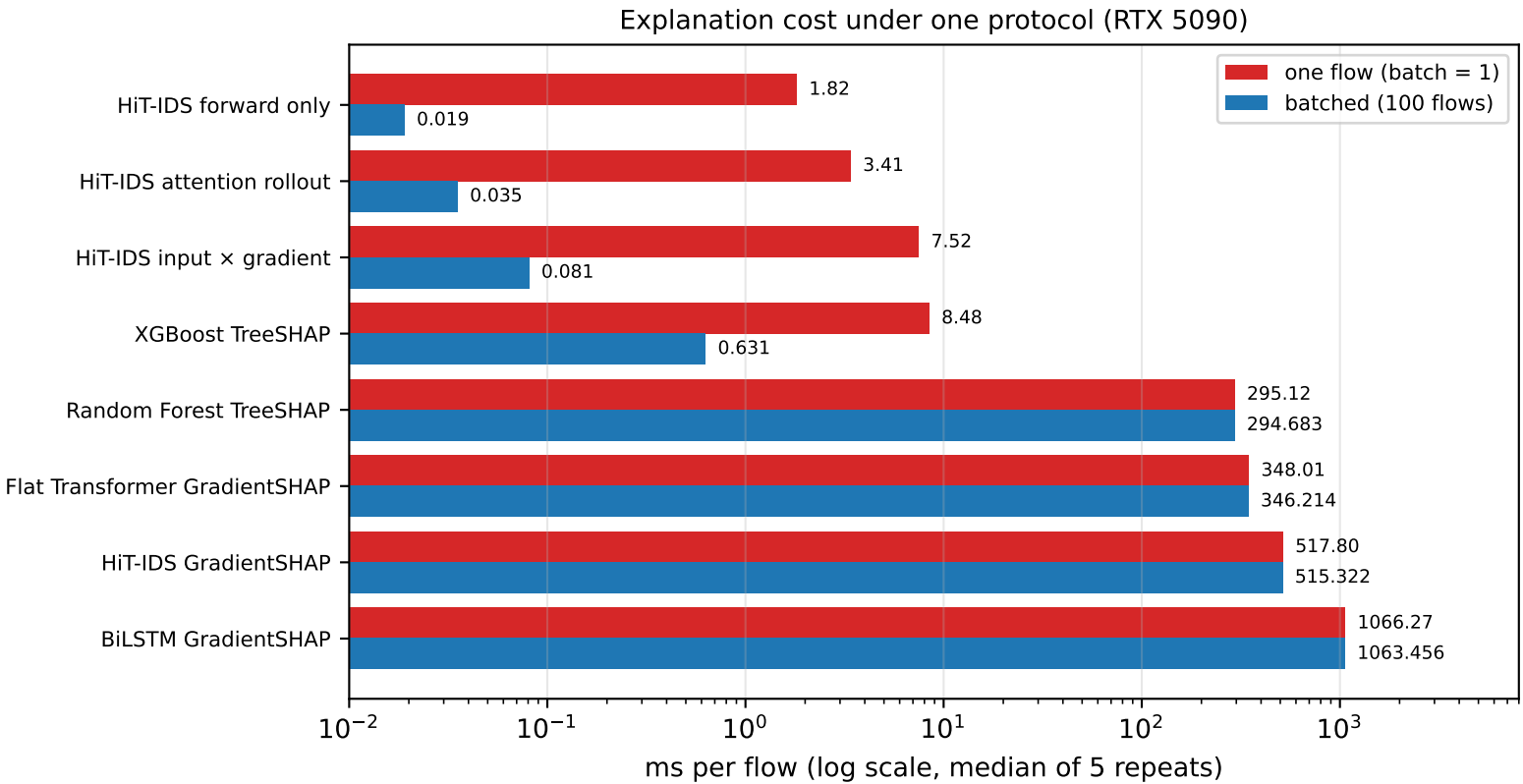
Explanation cost under one protocol (RTX 5090)
one flow (batch = 1)
batched (100 flows)
HiT-IDS forward only
1.82
0.019
HiT-IDS attention rollout
3.41
0.035
HiT-IDS input × gradient
7.52
0.081
XGBoost TreeSHAP
8.48
0.631
Random Forest TreeSHAP
295.12
294.683
Flat Transformer GradientSHAP
348.01
346.214
HiT-IDS GradientSHAP
517.80
515.322
BiLSTM GradientSHAP
1066.27
1063.456
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
ms per flow (log scale, median of 5 repeats)